

WEB TECHNOLOGIES

CSE-4004



LECTURE 8: Client-side Form Validations



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Client Side Form Validation

- It is the process of validating user input using HTML and JavaScript before it is sent to the server.

Data Validation

Data validation is the process of ensuring that user input is clean, correct, and useful.

- Validation can be performed in two ways:
- **Server side validation** is performed by a web server, after input has been sent to the server.
- **Client side validation** is performed by a web browser, before input is sent to a web server.

Why Client Side Validation?

- **Prevents submitting empty or wrong data**
- **Reduces load on server**
- **Makes form user friendly**

Types of of Client Side Validations

- 1. HTML5 Validation**
- 2. JavaScript Validation**

1. HTML5 Validation

- **required:** Field cannot be empty
- **type="email":** Checks email format
- **minlength** = Minimum characters
- **maxlength** = Maximum characters
- **min / max** : Number range
- **pattern:** Regular expression pattern
- **placeholder=** Shows hint text inside the input field.

required Attribute

- `<input type="text" required placeholder="Enter your name">`
- `<input type="email" required placeholder="Enter email">`
- `<input type="password" minlength="6" required>`

- `<input type="text" maxlength="10" placeholder="Max 10 characters">`
- `<input type="number" min="18" max="60" required>`

Regular Expression:

- `<input type="text" pattern="[0-9]{10}" placeholder="10 digit number" required>`
- `<input type="url" placeholder="Enter website URL">`
- `<input type="date" required>`

"checkbox" with required

- `<input type="checkbox" required>` I agree to terms
- `<input type="text" name="username" autocomplete="on">`
displays previously saved or entered data as suggestion.

```
<!DOCTYPE html>
<html>
<head>
  <title>Registration Form</title>
</head>
<body>
<center>
<h2>Registration Form</h2>
<form>
  <label for="name">Name:</label>
  <input type="text" id="name"
name="name" required
placeholder="Enter name">
```

```
<br> <br>
<label for="email"> Email:</label>
<input type="email" id="email"
name="email" required
placeholder="example@gmail.com">
<br> <br>
<br> <br>
  <label for="password">
    Password:</label>
    <input type="password"
id="password" name="password"
minlength="6" required
placeholder="Min 6 characters">
```

```
<br> <br>
<label for="age">Age:</label>
<input type="number" id="age"
name="age" min="18" max="60"
required>
<br> <br>
<label for="mobile">Mobile
Number:</label>
<input type="text" id="mobile"
name="mobile" pattern="[6-9][0-9]{9}"
placeholder="10-digit mobile
number" required>
<br> <br>
```

```
<label>Gender:</label>
<input type="radio"
name="gender" required> Male
<input type="radio"
name="gender"> Female
<br> <br>
<input type="checkbox"
id="terms" required>
<label for="terms">I agree to
the terms and conditions
</label>
<br> <br>
```

```
<button type="submit">Register</button>  
</form>  
</center>  
</body>  
</html>
```

Example

Registration Form

Name:

Email:

Password:

Age:

Mobile Number:

Gender: ☐ Male ☐ Female

☐ I agree to the terms and conditions

Regular Expression

- It is a sequence of characters that forms a search pattern.
- It is used for data validation, searching, and replacing specific text pattern in string.
- **Syntax of R.E : /pattern/**
- **let regex = /hello/;**
- **let regex = new RegExp("CSE"); //using constructor**

Regular Expression Methods

R.E Methods	Example
test()	returns true/false let r = /pen/; r.test("I have a Pen"); returns false let r = /pen/i; r.test("I have a Pen"); returns true
match()	returns matched string "20BCE".match(/\d+/); // ["20"]
replace()	"hello world".replace(/world/, "JS"); // "hello JS"
search()	Returns position of match "hello world".search(/world/); // 6



Regular Expression Methods

Special Symbols	Meaning
*	0 or more occurrences
+	1 or more occurrences
?	0 or one occurrences
{3}	Exactly 3
{2,5}	Between 2 & 5
g	Matches pattern globally.
i	Case sensitive
m	multiline
[]	Matches characters enclosed in []
^	Matches character at the beginning of line
\$	Matches characters at the end of line
()	Grouping R.Es example (RE1 RE2)

Example Patterns	Description
<code>"123abc4567".match(/\d{3}/g);</code>	<code>["123", "456"]</code>
<code>/[a-zA-Z]/</code>	<code>/[a-zA-Z]/.test("CSE");</code>
<code>/^[6-9]\d{9}\$/</code>	Match exactly 10 digits, starting with 6–9, nothing before or after.
<code>.</code>	Matches any character, <code>/a.c/</code> → matches abc, a c, a@c
<code>^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@\$!%*?&])[A-Za-z\d@\$!%*?&]{8,}\$</code>	At least 1 lowercase At least 1 uppercase At least 1 number At least 1 special character Minimum 8 characters

```
<!DOCTYPE html>
<html>
<head>
  <title>Form Validation with Alert</title>
</head>
<body>
  <h2>Registration Form</h2>

  <form id="form" onsubmit="return validateForm()">
    Username: <input type="text" id="username"><br><br>
    Email: <input type="text" id="email"><br><br>
    Password: <input type="password" id="password"><br><br>
    <input type="submit" value="Submit">
  </form>
```

```
<script>
```

```
function validateForm() {  
  const username = document.getElementById('username').value.trim();  
  const email = document.getElementById('email').value.trim();  
  const password = document.getElementById('password').value;  
  
  // Username validation: 3-15 letters or numbers  
  const usernamePattern = /^[A-Za-z0-9]{3,15}$/;  
  if(username === '')  
  {  
    alert('Username is required');  
    return false; // stop form submission  
  } else if(!usernamePattern.test(username))  
  {  
    alert('Username must be 3-15 letters or numbers');  
    return false;  
  }
```

// Email validation

```
const emailPattern = /^[a-zA-Z0-9._%+-]+@[a-z0-9.-]+\.[a-z]{2,4}$/;  
if(email === '') {  
  alert('Email is required');  
  return false;  
} else if(!emailPattern.test(email)) {  
  alert('Enter a valid email');  
  return false;  
}
```

//Password validation: min 8 chars, 1 uppercase, 1 lowercase, 1 number

```
const passPattern = /^(?=.*[a-z])(?=.*[A-Z])(?=.*\d){8,}$;/
if(password === '') {
  alert('Password is required');
  return false;
} else if(!passPattern.test(password)) {
  alert('Password must be at least 8 characters, include 1
uppercase, 1 lowercase, and 1 number');
  return false;
}
alert('Form submitted successfully!');
return true; // allow submission
} </script> </body> </html>
```

[Example](#)